Mathematics - Graduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Let A be the set of all ordered pairs of integers (m, n) such that 7 m + 12 n = 22. What is the greatest negative number in the set B = {m + n : (m, n) ∈ A}?

   (a) -5

   (b) 0

   (c) -3

   (d) -7

   (e) -4

2. A model of a park was built on a scale of 1.5 centimeters to 50 meters. If the distance between two trees in the park is 150 meters, what is this distance on the model?

   (a) 6 centimeters

   (b) 150 centimeters

   (c) 0.5 centimeter

   (d) 9 centimeters

   (e) 225 centimeters

3. Let V and W be 4-dimensional subspaces of a 7-dimensional vector space X. Which of the following CANNOT be the dimension of the subspace V intersect W?

   (a) 0

   (b) 1

   (c) 9

   (d) 8

   (e) 2

4. Solve 13 over 4 = x over 7.

   (a) 22.25

   (b) 23.25

   (c) 25.25

   (d) 23.75

   (e) 21.75

5. If all the values of a data set are the same, all of the following must equal zero except for which one?

(a) Mean
(b) Standard deviation
(c) Variance
(d) Interquartile range
(e) Quartile deviation

True / False (10 marks)

Answer True or False

6. True or False: The correct answer to 'suppose $u = \arctan \frac{y}{x}$, what is numeric of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$?' is '-0.5'.

7. True or False: The arc length of the curve y = (1/4)x^4 over the interval [1,2] using the Trapezoidal Rule T_5 is approximately 4.322.

8. True or False: The simplified form of the fraction $\frac{4}{\sqrt{108}+2\sqrt{12}+2\sqrt{27}} is \frac{1}{4\sqrt{27}}$$.

9. True or False: Every group of order 159 is cyclic, implying it has no nontrivial subgroups.

10. True or False: The limit of 3 n(a_$n)^3$ as n approaches infinity is equal to 1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the value of the integral $\int_{|z|=r} \frac{f(z)}{(z-z_1)(z-z_2)} dz$ for a bounded entire function f and points z_1, z_2 in the ball $B(0, r)$. Justify your reasoning.

12. Determine the value of N such that the equation $\frac{4^{(3^3)}}{(4^3)^3} = 2^N$ holds true. Explain your reasoning and the steps taken to arrive at your answer.

End of Paper